

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – [Agentic AI for Mental Health Awareness & Suicide Prevention]

Domain of Project – [Healthcare]

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Problem Statement -

Problem Statement : Agentic AI for Mental Health Awareness & Suicide Prevention

The Challenge

Mental health issues such as depression, anxiety, and suicidal thoughts are growing worldwide. Stigma, lack of awareness, and limited access to timely help prevent many individuals from seeking support. Traditional systems are often reactive, leaving gaps in early detection and preventive care.

The Objective

Build an Agentic AI-powered Mental Health Awareness & Suicide Prevention Agent that is empathetic, proactive, and safe. The solution should:

Promote Awareness & Education – Share accessible resources on mindfulness, coping strategies, and self-care.

Enable Early Detection (RAG) – Use Retrieval-Augmented Generation to identify distress signals in conversations, journaling inputs, or safe online data.

Provide Empathetic Support – Build a conversational agent that listens actively and responds with empathy (with clear disclaimers: not a substitute for professional medical advice).

Predict & Prevent Risks – Forecast stress peaks, suggest preventive strategies, and issue alerts when high-risk signals are detected (with consent).

Connect to Human Support – Link users to helplines, counsellors, or community support for timely intervention.

Proposed Solution -

Proposed Solution: Agentic AI for Mental Health Awareness & Suicide Prevention – SERENE — A Safe Space to Be Heard.

Serene is an **Agentic AI-powered Mental Health Awareness & Suicide Prevention system** designed to provide early, empathetic, and proactive support.

The system allows users to share their **mood, journal entries, and conversations** voluntarily. Specialized AI agents analyse these inputs to identify potential signs of **stress, emotional distress, and high-risk situations** without attempting to provide a medical diagnosis.

- **RAG-Based Support** – Retrieves information from trusted mental-health resources to provide reliable guidance on mindfulness, coping strategies, self-care, and professional help.
- **Empathetic AI Agent** – Provides personalized, non-judgmental conversations while clearly communicating that AI is not a replacement for professional care.
- **Early Risk Detection** – Identifies changes in emotional patterns and classifies the level of concern as **Low, Moderate, High, or Immediate**.
- **Preventive Intervention** – Suggests suitable coping techniques and encourages users to seek support before distress escalates.
- **Human Support & Escalation** – For serious safety concerns, connects users with **helplines, counsellors, trusted contacts, or emergency services**, keeping human intervention at the center of crisis support.
- The solution is developed using **IBM Bob, IBM Langflow/Orchestrate, IBM AI models, RAG, and a vector database**, creating a **safe, intelligent, personalized, and proactive mental-health support system rather than a conventional chatbot**.

Technology Used

- **IBM Bob platform** -For developing and orchestrating the Agentic AI mental-health application
- **RAG**– For retrieving trusted mental-health resources and generating evidence-grounded responses.
- **Vector Database** (FAISS/Chroma) – For efficient storage and retrieval of mental-health knowledge.

IBM BOB code output

The screenshot displays the IBM BOB development environment. The left sidebar shows the Explorer view with the project structure. The main editor window shows the code for `crisis_detector.py`. The code defines a `CrisisLevel` enum and a `CrisisAssessment` dataclass. The `CRITICAL_KEYWORDS` list contains various phrases related to crisis and suicide.

```
1 """
2 Crisis Detection Agent
3 Analyzes user messages for signs of distress, crisis, and suicidal ideation.
4 Uses a multi-level risk assessment approach.
5 """
6
7 import re
8 from dataclasses import dataclass
9 from enum import Enum
10 from typing import List, Tuple
11
12
13 class CrisisLevel(Enum):
14     NONE = 0
15     LOW = 1
16     MODERATE = 2
17     HIGH = 3
18     CRITICAL = 4
19
20
21 @dataclass
22 class CrisisAssessment:
23     level: CrisisLevel
24     score: float
25     triggers: List[str]
26     requires_immediate_action: bool
27     recommended_response_type: str
28     safety_resources_needed: bool
29
30
31 # — Keyword banks —
32
33 CRITICAL_KEYWORDS: list[str] = [
34     "want to die", "kill myself", "end my life", "suicide", "suicidal",
35     "don't want to live", "no reason to live", "better off dead",
36     "take my own life", "end it all", "can't go on", "not worth living",
37     "planning to die", "goodbye forever", "final goodbye", "last day",
```

The right sidebar shows the IBM BOB interface with a task list and a "To Run" section.

Build an Agentic... 62.5k / 270.0k 3.63

Build the Flask backend with Gr... 2/6

- AFSP/SAMHSA/WHO safe messaging guidelines hardcoded in the system prompt
- Real-time crisis indicator in the header changes color/label live

Wellness Tools (UI Modals)

- Animated Breathing Exercise — Box breathing with real-time circle animation, countdown
- Grounding Technique — 5-4-3-2-1 step-by-step guide
- Crisis Hotlines — All 8 hotlines with urgency-tier categorization

To Run

```
cd mental_health_agent
pip install -r requirements.txt
python app.py
# Open: http://127.0.0.1:5000
```

Or double-click `run.bat` on Windows.

12 files changed Undo all Show all

Follow up or start new task Ctrl+N

+ Agent Permissions

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IBM PROJECT

Upgrade your plan

EXPLORER

IBM PROJECT

- mental_health_agent
 - __pycache__
 - agents
 - __pycache__
 - __init__.py
 - crisis_detector.py
 - empathy_engine.py
 - resource_agent.py
 - wellness_agent.py
 - data
 - static
 - templates
 - index.html
- app.py
- config.env
- requirements.txt
- run.bat

OUTLINE

TIMELINE

PROJECTS

index.html

mental_health_agent > templates > index.html

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6   <title>Serene – Mental Health Support Agent</title>
7   <link rel="stylesheet" href="/static/css/style.css" />
8   <link rel="preconnect" href="https://fonts.googleapis.com" />
9   <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&family=L
10 </head>
11 <body>
12
13 <!-- EMERGENCY BANNER -->
14 <div id="emergency-banner" class="emergency-banner hidden">
15   <div class="emergency-inner">
16     <span class="emergency-icon">🚨</span>
17     <span id="emergency-text"></span>
18     <button onclick="dismissBanner()" class="banner-close">X</button>
19   </div>
20 </div>
21
22 <!-- SIDEBAR -->
23 <div class="sidebar" id="sidebar">
24   <div class="sidebar-header">
25     <div class="logo-area">
26       <div class="logo-icon">🌿</div>
27       <div>
28         <div class="logo-name">Serene</div>
29         <div class="logo-tagline">Mental Health Companion</div>
30       </div>
31     </div>
32     <button class="sidebar-toggle-btn" id="sidebar-toggle" onclick="toggleSidebar()">☰</button>
33   </div>
34
35   <div class="sidebar-section">
36     <button class="new-chat-btn" onclick="startNewChat()">
37       <span>👤</span> New Conversation
```

IBM BOB

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New task

12 files changed Undo all Show all

Follow up or start new task Ctrl+I

+ Agent Permissions

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Search

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FileEditSelectionViewGoRunTerminalHelp

IBM PROJECT

Upgrade your plan

EXPLORER

IBM PROJECT

mental_health_agent

__pycache__

agents

__pycache__

__init__.py

crisis_detector.py

empathy_engine.py

resource_agent.py

wellness_agent.py

data

static

templates

index.html

app.py

config.env

requirements.txt

run.bat

OUTLINE

TIMELINE

PROJECTS

app.py

mental_health_agent > app.py > ...

```
1 """
2 Mental Health Awareness & Suicide Prevention Agent
3 Main Flask Application
4 Powered by Groq API + Agentic AI Architecture
5 """
6
7 import os
8 import json
9 import uuid
10 import logging
11 from datetime import datetime
12 from typing import List, Dict, Any
13
14 from flask import Flask, request, jsonify, render_template, session
15 from flask_cors import CORS
16 from groq import Groq
17 from dotenv import load_dotenv
18
19 # Load env from config.env
20 load_dotenv(dotenv_path="config.env")
21
22 # — Agent imports —
23 from agents.crisis_detector import CrisisDetector, CrisisLevel
24 from agents.empathy_engine import EmpathyEngine
25 from agents.resource_agent import ResourceAgent
26 from agents.wellness_agent import ProactiveWellnessAgent
27
28 # — App setup —
29 app: Flask = Flask(import_name=__name__)
30 app.secret_key = os.getenv("FLASK_SECRET_KEY", "mh_secure_key_default")
31 CORS(app, supports_credentials=True)
32
33 logging.basicConfig(
34     level=logging.INFO,
35     format="%(asctime)s [%(levelname)s] %(message)s",
36     datefmt="%Y-%m-%d %H:%M:%S",
37 )
```

IBM BOB

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12 files changed Undo all Show all

Follow up or start new task Ctrl+N

+ Agent Permissions

Indexing completed. Java: Ready Bob Findings

Ln 1, Col 1 Spaces: 4 UTF-8 LF Python Python 13.13 Bob Settings

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Search

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Final Output

**Serene**
MENTAL HEALTH
COMPANION

New Conversation

QUICK SUPPORT

I feel anxious

I feel depressed

I feel alone

I'm stressed

I need urgent help

WELLNESS TOOLS

Breathing Exercise

Crisis Hotlines

Grounding Technique

Important: Serene is an AI companion and is not a substitute for professional mental health care. If you are in crisis, please contact a crisis line or seek help from a healthcare professional.

**Serene**
Here for you · Always available

Feeling okay



Welcome to Serene

A safe, compassionate space to talk about how you're feeling.

I'm here to listen — without judgment, without rush. Whether you're feeling anxious, sad, overwhelmed, or you just need someone to talk to, I'm here.



Start talking



Calm exercise



Crisis lines

 This conversation is private. You don't need to share your name.

Share what's on your mind... I'm here to listen.



0 / 2000

Press Enter to send · Shift+Enter for new line

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 Search



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10-09-2026

Bob — ×New tab ×Generat ×Edunet ×API Key ×API Key ×IBMDesi ×Comple ×GitHub ×Serene ×+Ask Gemini

←→↻127.0.0.1:5000☆🔍G⋮

🧘 Breathing Exercise

×

Box breathing activates your calm nervous system. Find a comfortable position and follow along.

Ready

4

➊ Inhale (4s)

➋ Hold (4s)

➌ Exhale (4s)

➍ Hold (4s)

Cycle 0 of 4

Start

Stop

29°

🏠🖨️🔍 Search

🌐📁📅🔍?📧📱3📺🌐🔍📅📅📅📅

^🌐ENG IN📶🔊🔋

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**Serene**

Here for you · Always available

Feeling okay

I've been feeling very sad and depressed lately

04:13 PM



I hear how heavy things feel for you right now, and I'm really sorry you're carrying so much sadness. It sounds like you've been in a low place for a while, and that can feel overwhelming and isolating. I'm here with you, ready to listen and to help you explore ways to find a little relief and some hope.

🌱 A few things that might help you right now

| What it is | How it can help | Simple ways to try it |

|-----|-----|-----|

| **Validate your feelings** | Acknowledging that the sadness is real can reduce the pressure to "fix" it right away. | Write a short note to yourself: "I'm feeling sad, and that's okay. I'm allowed to feel this." |

| **Grounding or breathing exercises** | Helps calm the nervous system when thoughts feel overwhelming. | Try the 4-7-8 breath: inhale for 4 seconds, hold for 7, exhale for 8. Repeat a few times. |

| **Small, doable actions** | Tiny steps can create a sense of agency without feeling like a huge demand. | Choose one tiny task (e.g., get a glass of water,

Share what's on your mind... I'm here to listen.



0 / 2000

Press Enter to send · Shift+Enter for new line

This technique brings you back to the present moment. Go slowly through each step.

4 THINGS YOU CAN TOUCH

Notice 4 physical sensations — the chair beneath you, the floor under your feet...

2 THINGS YOU CAN SMELL

Take a breath. Notice 2 scents — the air, your clothes, something nearby...

1 THING YOU CAN TASTE

Notice 1 taste in your mouth right now.

Role of Agentic AI in the solution

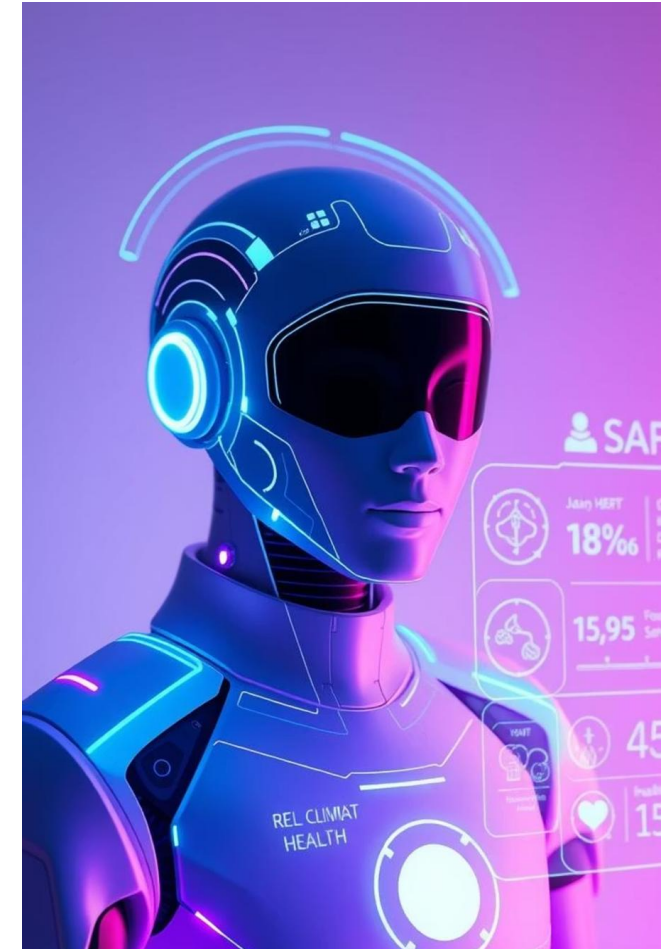
Role of Agentic AI in “Serene”

Agentic AI enables **Serene** to function as an intelligent, proactive system rather than a simple chatbot. Using **IBM AI models**, it coordinates multiple specialized agents to provide personalized and safety-focused mental-health support. Each agent performs a specific role—such as **understanding conversations**, **retrieving trusted mental-health information**, **detecting distress signals**, **suggesting preventive strategies**, and **connecting users to human support**—while working collaboratively. This **multi-agent approach** enables better decision-making and timely intervention.

Agentic AI provides **context awareness**, understanding the user's current emotional state and previous consented interactions to deliver relevant responses. It also supports **proactive intervention** by identifying increasing distress patterns and recommending appropriate support before a situation escalates.

Additionally, agents communicate with each other through an orchestrated workflow, while a **Safety Agent** ensures that responses remain responsible and do not provide harmful guidance.

Overall, Agentic AI makes Serene **empathetic, proactive, personalized, and safety-focused**, acting as an intelligent support system that helps users access the right guidance and human assistance when needed.



Novelty and Uniqueness

The proposed **Mental Health Support Agent** stands out by combining **Agentic AI, RAG, early risk detection, personalization, and human-in-the-loop support** into a single intelligent and safety-focused system.

- **Multi-Agent Intelligence** – Instead of relying on a single chatbot, multiple specialized agents collaborate to provide empathetic, context-aware, and safety-focused mental health support.
- **RAG-Based Reliability** – Retrieves information from trusted mental-health resources before generating responses, reducing misinformation and improving the reliability of guidance.
- **Early Distress Detection** – Identifies potential emotional distress and concerning signals from voluntary conversations, journal entries, and mood check-ins for timely intervention.
- **Personalized Prevention** – Provides personalized coping strategies, mindfulness activities, self-care suggestions, and support recommendations based on the user's situation and changing needs.
- **Proactive Risk Monitoring** – Analyses consented mood and interaction patterns to identify increasing distress and recommend preventive action before the situation escalates.
- **Human-in-the-Loop Safety** – Goes beyond AI support by connecting users with **counsellors, helplines, trusted contacts, or emergency services** when serious safety concerns arise.
- **Safety & Privacy First** – Uses consent-based monitoring, AI safety checks, privacy-aware data handling, and clear disclaimers to ensure responsible mental-health assistance.
- **IBM-Powered & Scalable** – Built using **IBM Bob, IBM Langflow/Orchestrate, and IBM foundation models**, enabling an enterprise-oriented, governed, and scalable solution.
- This makes the solution a **smart, empathetic, adaptive, and proactive mental-health support system**, rather than just a conventional AI chatbot.

Git Hub Link

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Github URL –<https://github.com/gbhargavi9105/IBM-Certification-and-project>

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Future Scope

1. Integration with Wearables & Health Devices –

The system can be integrated with **smartwatches and health devices** to collect consented data such as sleep patterns, heart rate, activity levels, and other wellness indicators. This can help identify changes in well-being and enable more personalized and timely mental-health support.

2. AI-Powered Voice Assistant & Multilingual Support –

Adding a **voice-enabled assistant and multilingual support** will improve accessibility and make the system easier to use. Users will be able to communicate naturally in their preferred language, making the solution more inclusive for diverse populations.

3. Advanced Predictive Risk Analysis –

Future versions can use **long-term, consented mood and behavioural trends** to identify increasing distress earlier and provide preventive interventions before a crisis develops.

4. Professional & Emergency Support Integration –

The platform can be connected with **counsellors, mental-health professionals, helplines, and emergency services** to enable faster human intervention when serious safety concerns are detected.

5. Personalized Mental Wellness Plans –

The system can generate personalized **daily wellness plans, mindfulness activities, coping strategies, and follow-up reminders** based on the user's needs and preferences.

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



G. Bhargavi

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



Issued on: Sep 08, 2026

Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/891da984-c15f-4d78-b9f1-909234b0ef23>



Certificates Earned:

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

G. Bhargavi

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 09 Sep 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.